

MiniLabel v1 – Simple Specification

Goal: Build a very small, internal annotation tool (Label-Studio-lite) for students. Focus on learning, not completeness.

Supported tasks (v1 only)

- Text classification (single label)
- Named Entity Recognition (character spans)
- Image object detection (bounding boxes)

Non-goals

- No segmentation masks or polygons
- No multi-user workflows
- No consensus, review, or analytics

Minimal stack

- Backend: Python + FastAPI
- Database: SQLite
- Frontend: Simple HTML + JS OR Streamlit
- Drawing boxes: HTML Canvas

Core concepts

- Project: defines task type and label list
- Item: one text or one image
- Annotation: JSON saved per item

Annotation formats

Text classification:

```
{ "label": "NEG" }
```

NER:

```
{ "spans": [ { "start": 10, "end": 18, "label": "ORG" } ] }
```

Bounding box (normalized coordinates):

```
{ "label": "kiln", "bbox": { "x": 0.1, "y": 0.2, "w": 0.3, "h": 0.4 } }
```

API (minimum)

- GET /items – list items
- GET /items/{id} – fetch item
- PUT /items/{id}/annotation – save JSON annotation

Milestone plan

- Week 1: FastAPI + SQLite + item loading
- Week 2: Text classification + NER UI
- Week 3: Image bbox UI + JSON export

Success criterion: students can annotate data and export clean JSON without crashes.